

Cybersecurity

Ports, Protocols, and the OSI Model

Lesson 8.2



Learning Objectives

By the end of today's lesson, you will be able to:

- 1 Interpret data in network packets by analyzing their headers, payloads, and trailers.
- 2 Explain the role of ports in specifying a network packet's destination.
- 3 Associate common protocols with their assigned ports.
- 4 Explain how encapsulation and decapsulation let different protocols interact with one another.
- 5 Use the layers of the OSI model to identify sources of problems on a network.
- 6 Capture and analyze live network traffic using Wireshark.



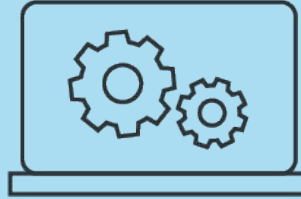
Welcome



Recap

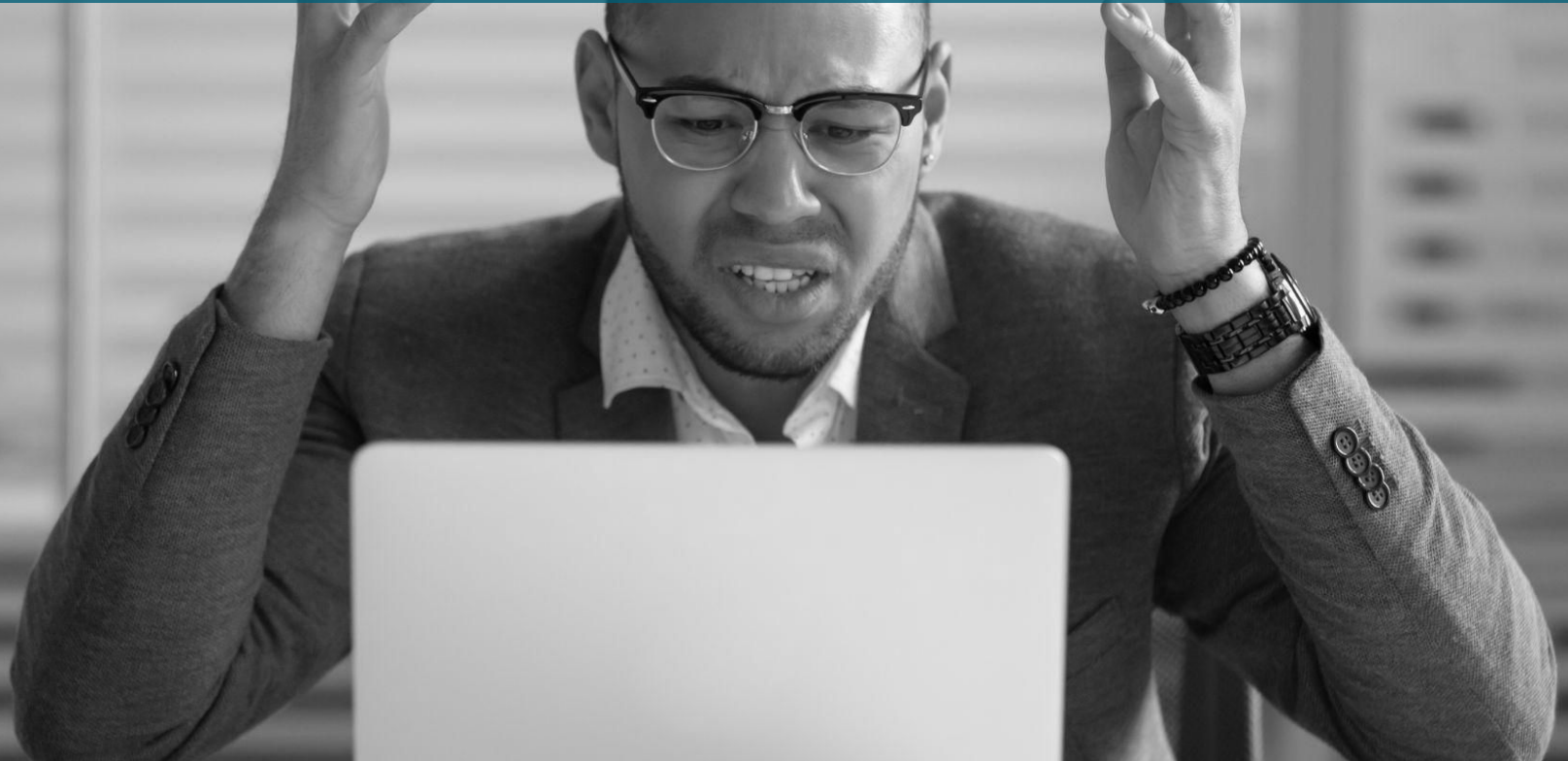
In the last lesson, we discussed how:

- 1 Networking's main function is to share data across entities.
- 2 Various network devices are used to facilitate data sharing.
- 3 IP and MAC addresses direct the path that data takes across entities.



Protocols

With so much going on behind the scenes in network communication, it's important to have systems in place to avoid the many potential errors.



Protocols

Behind-the-scenes actions



Opening and closing connections



Finding the location to send the data



Converting and deconverting the data



Transmitting the data reliably



Handling errors

Real-World Protocols

When military officers exchange messages over a radio, they can encounter the following issues:

- Communications don't come in clearly.
- Multiple communications are sent at the same time.
- Communications are cut off.



Real-World Protocols

Military personnel use communication protocols to ensure messages are transmitted, received, and understood clearly.

A black silhouette of a soldier in profile, facing right, wearing a helmet and holding a radio. A large white number '1' is on the back.

"This is Officer 1.
Are you there? **Over.**"

A black silhouette of a soldier in profile, facing left, wearing a helmet and holding a radio. A large cyan number '2' is on the back.

"**Roger**, this is Officer
2. I am here. **Over.**"

"Start the attack at
noon. **Over.**"

"**Roger! Out.**"

Real-World Protocols

Military officers avoid communication issues by using a fixed mode of communication.

01

“Over”

Signals the end of a specific line of communication.

02

“Roger”

Signals that the message was received completely.

03

“Out”

Signals that the message was received, the exchange is complete, and the order will be followed.

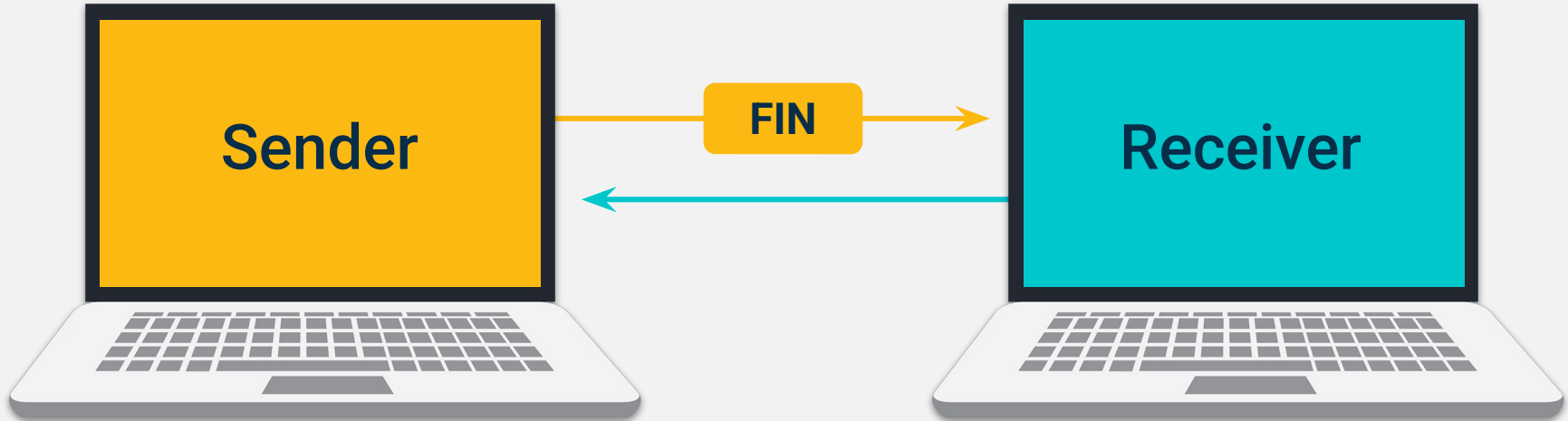


These strict rules are known as a **protocol**. They impose structure by specifying the precise meaning of keywords and where in a message they must appear.

Protocols and Networks

Networks use protocols to ensure messages are fully sent and understood.

Similar to the military's use of "over," a network uses the TCP message FIN to indicate the end of the transmission.



Some **protocols** you may be familiar with:



**Hypertext Transfer Protocol
(HTTP)**



**File Transfer Protocol
(FTP)**

Networking Protocols

Some other important protocols include:

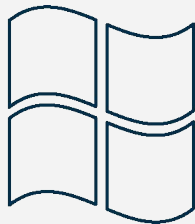
Password Authentication Protocol

PAP is used for authenticating users.



Server Message Block

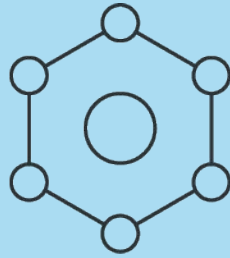
SMB is a Windows-based protocol for sharing files.



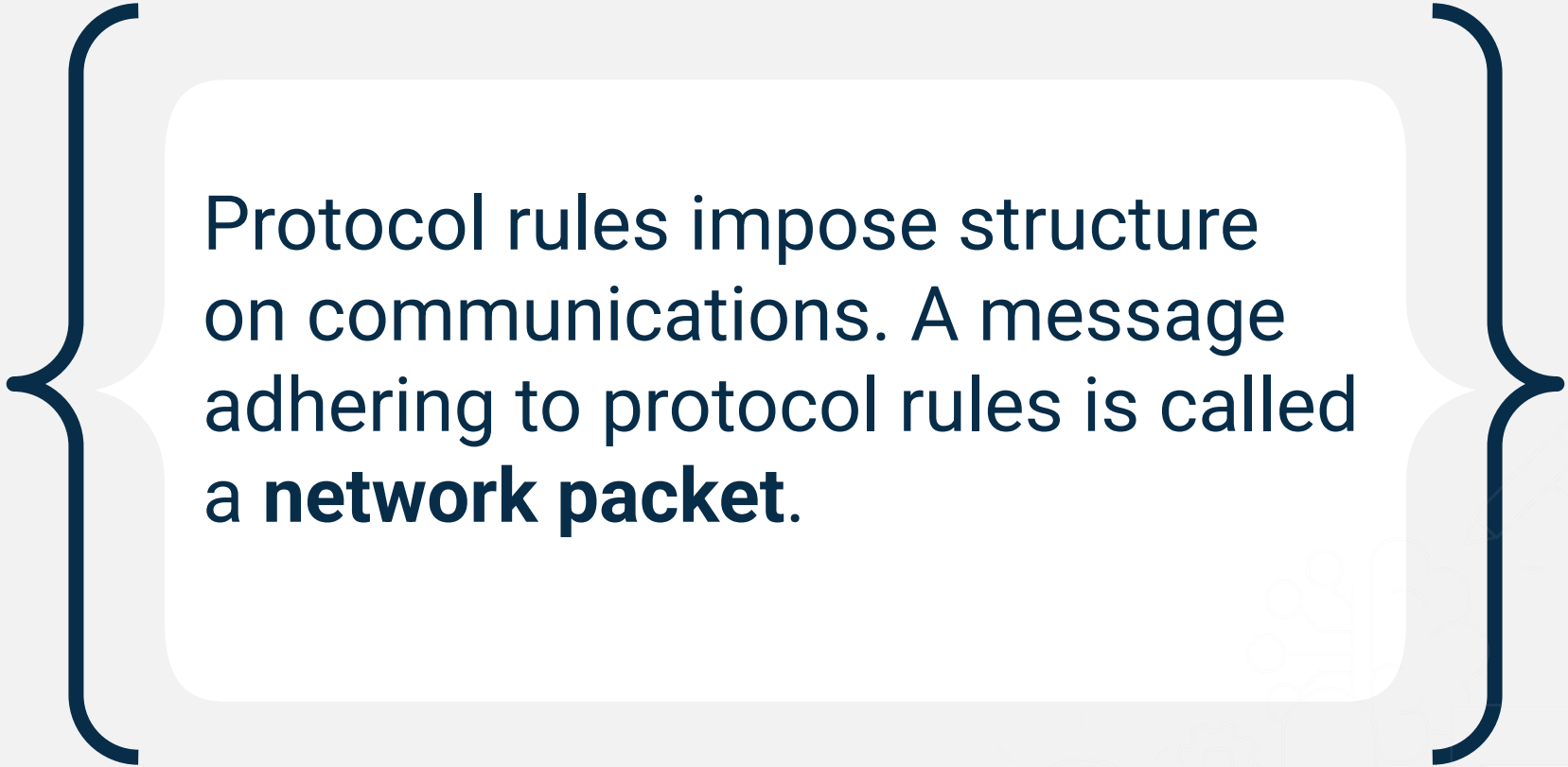
Network Basic Input/Output System

NetBIOS allows computers to communicate on a local network.





Network Packet Structure



Protocol rules impose structure on communications. A message adhering to protocol rules is called a **network packet**.

As we know from the previous lesson, the client and server communicate by exchanging binary data.



Network Packet Structure

The binary data is grouped into separate pieces, known as **packets**, and transmitted across the network.

Packet



Network Packet Structure

The **header** contains information about the data in the packet.

- Name
- Version
- Protocol
- Length



Network Packet Structure

The **payload** contains the data being delivered to the destination.

- **Body or Data**
- Message



- In the previous military radio example, the payloads are, "**I am here**," "**Start the attack at noon**," etc.

Network Packet Structure

The **trailer** informs the receiver that it has reached the end.

- Validation
- **Footer**
- Not always used





Each of these three sections is further subdivided into **fields**.



Packets

The header may contain **fields** such as:

Protocol	Specifies the name of the protocol in use.
Version	Specifies which version of the protocol is in use.
Destination Address	Specifies where the packet is going.
Length of Packet	Tells the receiver how much data is in the packet payload.



Each of these parts and the fields within them will appear in an agreed-upon order and size.

This **order is dictated** by the rules of the protocol in use.



Packets

Packet example

The **version field** is indicated in the header of an IP packet.

```
01000101000000000000000011101101000000111100011000100000000000000100000000000110
```

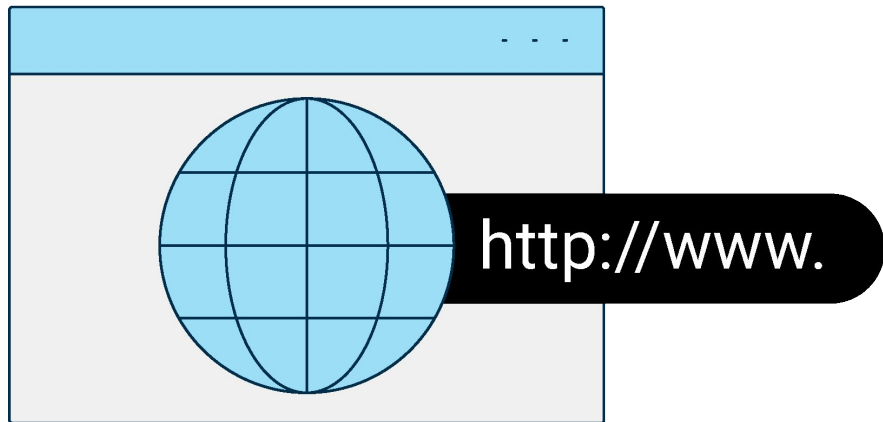
In binary, **0100** translates to 4.

Therefore, this field indicates that this header uses **Version 4**.



Packets

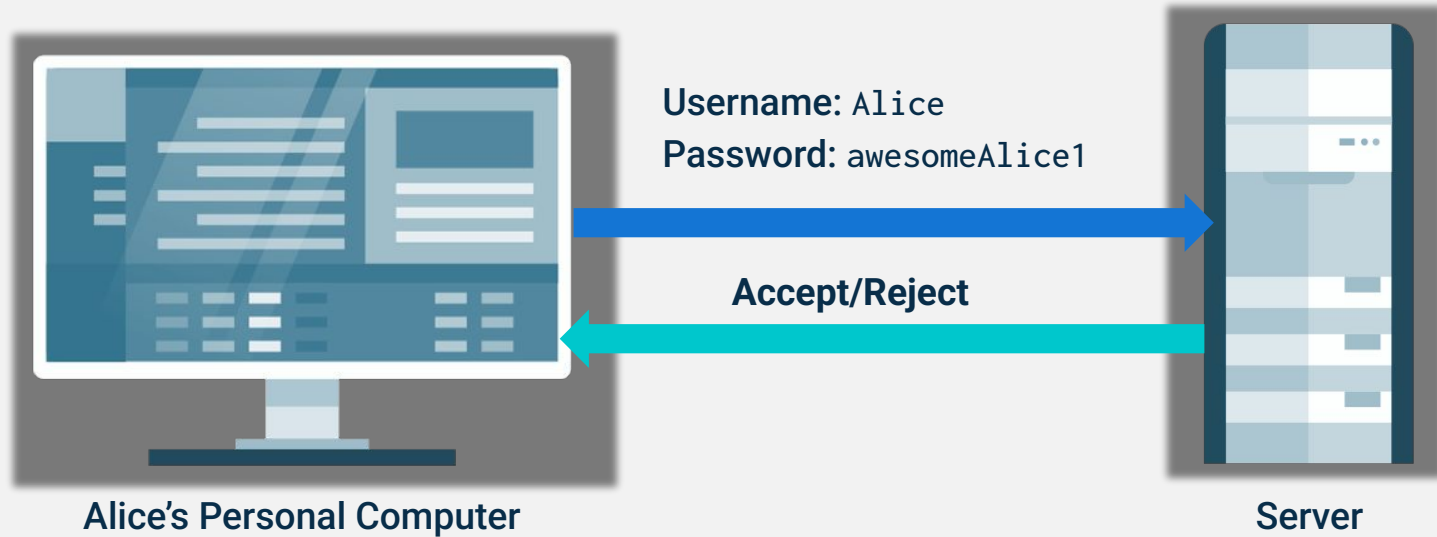
HTTP headers



- Different protocols contain different header and payload fields.
- For example, **HTTP messages** include a host field in their header.
- Since **FTP** does not involve websites, it does not have a host field.

Protocol Example: Authentication

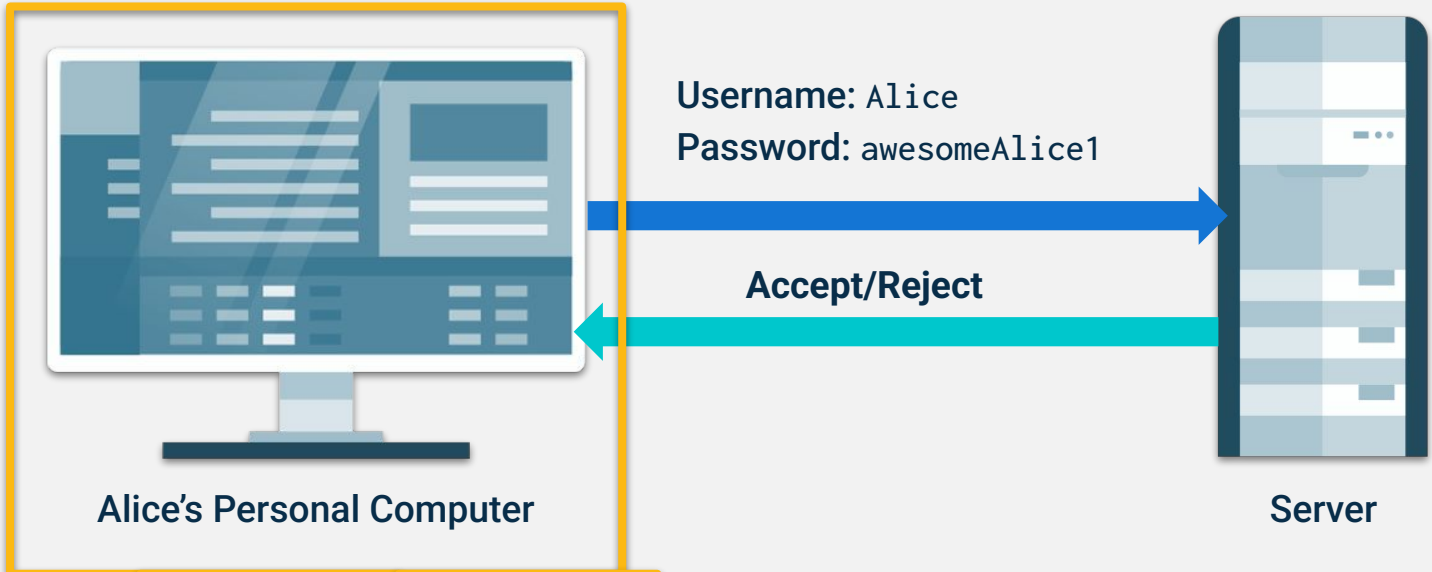
Password Authentication Protocol (PAP)



PAP two-way handshake

Protocol Example: Authentication

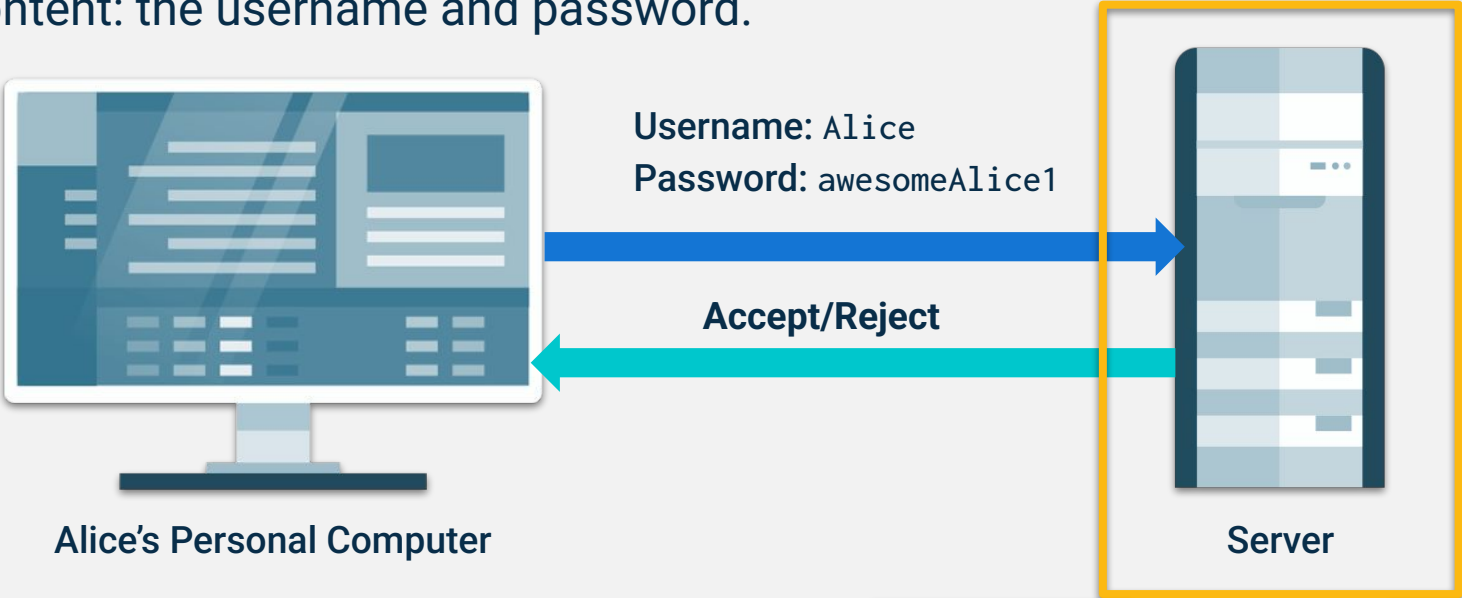
The client request contains bits in a specific order and length, per the standard and rules of the protocol.



1 Byte	1	2 Bytes	1 Byte	Variable	1 Byte	Variable
Code: 1	ID	Length	Username length	Username	Password length	Password

Protocol Example: Authentication

The server receiving the request will know where to look in the bitstream for content: the username and password.



1 Byte	1	2 Bytes	1 Byte	Variable	1 Byte	Variable
Code: 1	ID	Length	Username length	Username	Password length	Password



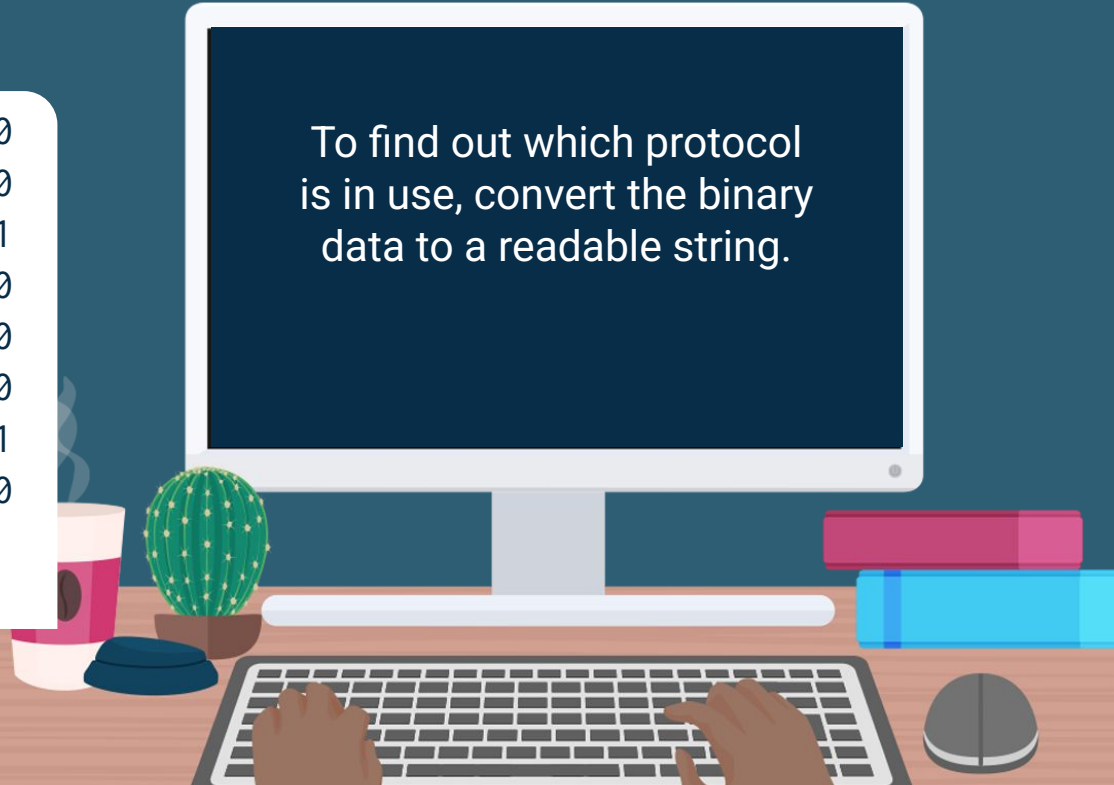
The most important fields in PAP are **sent-username**, which contains the username the user submitted, and **password**, which contains the password they sent.

Interpreting Protocols from Raw Binary Data

Security professionals rely on web tools to convert binary data and determine the protocol being used.

```
011100000110000101110000001000000  
111001101100101011011100111010000  
101101011101010111001101100101011  
100100110111001100001011011010110  
010100100000010100000100000101010  
000010101010101001101000101010100  
100010000001110000011000010111001  
101110011011101110110111101110010  
011001000010000000110111
```

To find out which protocol
is in use, convert the binary
data to a readable string.





Activity

Interpret Protocols Activity

In this activity, you will play the role of a security analyst at Acme Corp.

- Your manager suspects that one of your colleagues has been stealing important company data. In order to prevent the leak of trade secrets, your manager requests that you analyze several logs of the potentially rogue employee's recent network activity.
- Your task is to convert the raw binary data into a readable format and determine which protocol is being used.

Suggested Time
15 minutes

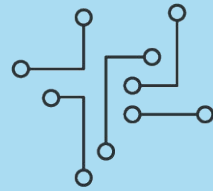


Time's Up!
Let's review



Questions?





Ports

Ports Real-world Analogy

Bob wants to show a video presentation to Alice, a coworker who works across town. Bob will be presenting from Room 33, the video conference room in his office building.

- Bob told Alice to meet him at his office, but only gave her the building address, 150 Main Street.
- Because Bob didn't specify which room he'd be in, Alice will be able to find the office building, but not Bob or the video presentation.





The address of the office building,
150 Main Street, is the **IP address**.

The video conference room,
Room 33, is the **port**.

Ports Real-World Analogy

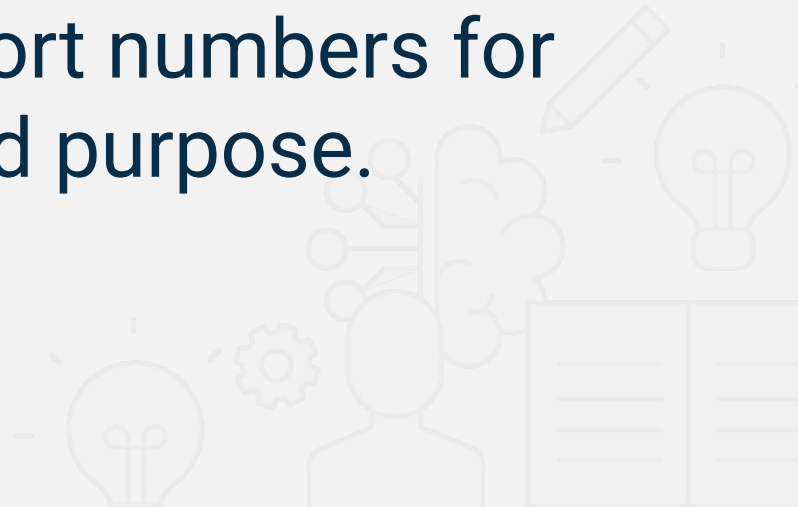
Ports are the access points for transmitting and receiving data.

- Ports are like doors that can be opened and closed.
- Since a port is considered an access point into a system, it is important that IT professionals do not allow unauthorized access to them.
- Unauthorized access can potentially lead to a breach.





Internet Assigned Numbers Authority (IANA) is the entity officially responsible for assigning port numbers for a designated purpose.

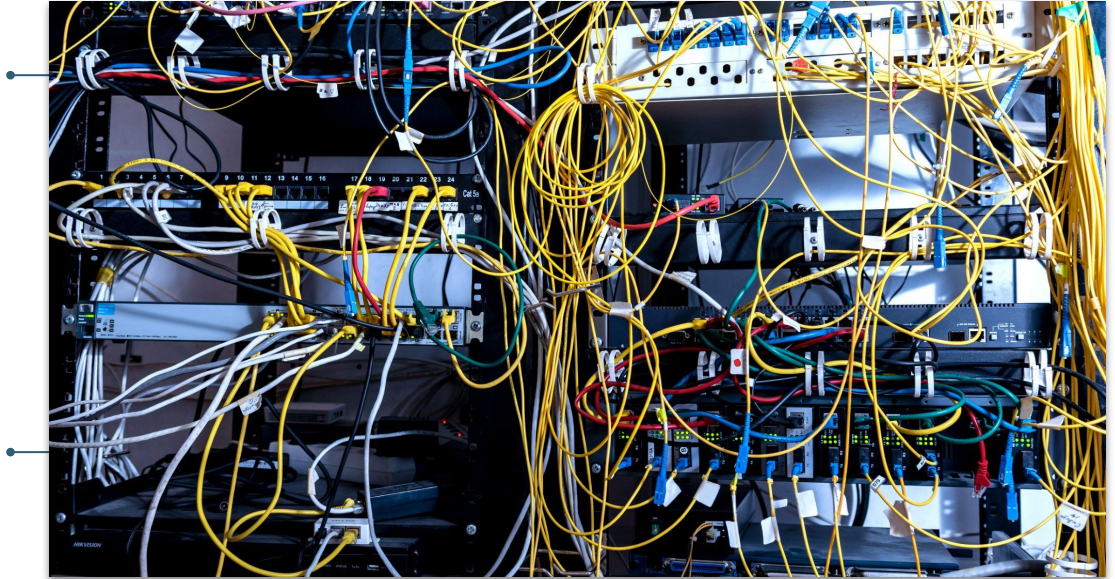


Virtual Ports

Computers do not have enough physical space for every protocol.

Every protocol is assigned a virtual port number.

The corresponding port is the **destination port**.



Port Numbers

There are **65,536** virtual ports numbered from 0 to 65535 divided into **3 ranges**:

System Ports

- Well-known ports
- 0-1023
- Restricted

Registered Ports

- User ports
- 1024–49151
- Restricted

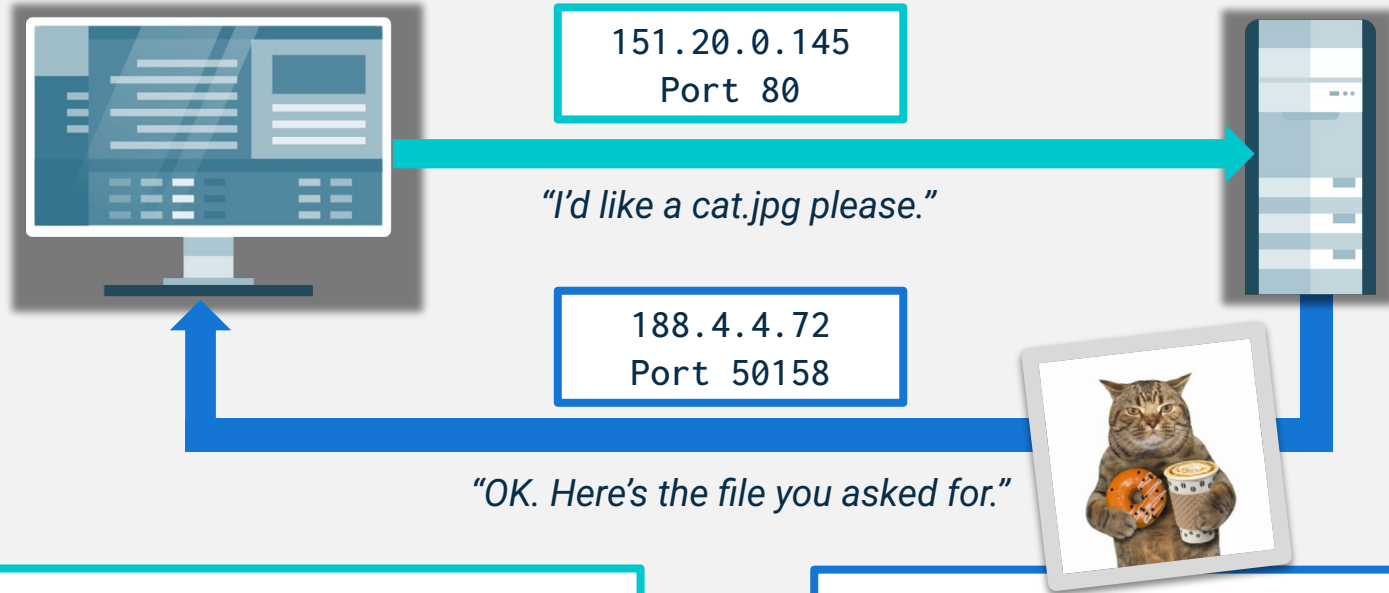
Dynamic Ports

- Private ports
- 49152–65535
- Restricted

Common Ports

Port 80	HTTP	Sends web traffic
Port 443	HTTPS	Sends encrypted web traffic
Port 21	FTP	Sends files
Port 22	SSH	Securely operates network services
Port 25	SMTP	Sends emails
Port 53	DNS	Translates domains into IP addresses

Source / Destination Ports



The client (initiator) has a **source** port. Source ports are randomly generated from the dynamic port range (49152 - 65535.)

The server (receiver) has a **destination** port depending on the protocol. Destination ports do not change, so we can associate a port with a certain protocol.



Let's Recap



Recap

Ports and protocols review

- 1 IP addresses are used to locate a computer on a network, such as the internet.
- 2 A port number is used to locate a specific service on that computer.
- 3 Both numbers are required to transport data.
- 4 Any service can be run on any port, but certain services have associated ports.



Questions?





Activity

Ports

In this activity, you will continue to play the role of a security analyst at Acme Corp.

- Your task is to determine the **source** and **destination port** for each request, as well as the protocol for each destination port.
- You must then research the protocol to determine what kind of activities the potentially rogue employee may be conducting.

Suggested Time
15 minutes



Time's Up!
Let's review

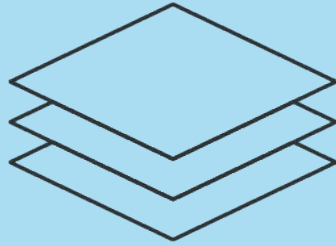


Questions?





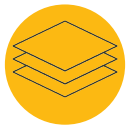
Break
15 mins



OSI Layers

An aerial night view of a city with a network overlay. The city lights are visible in the background, and a network of glowing blue lines and nodes is superimposed over the image. A large yellow circle is on the left side, containing text.

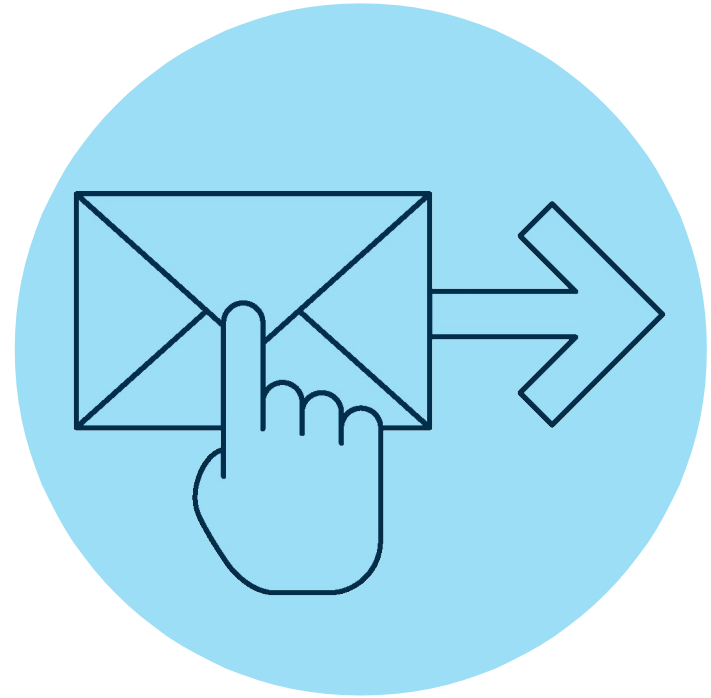
When data travels
across a network,
it goes through
multiple steps and
processes to reach
its destination.



OSI Layers

For example, the process of sending an email starts with the following steps:

1. Convert the text of the user's email into a format the email application can understand.
2. Add the destination address and destination port to this data.
3. Convert this packet to a format that can be transmitted through physical wires.

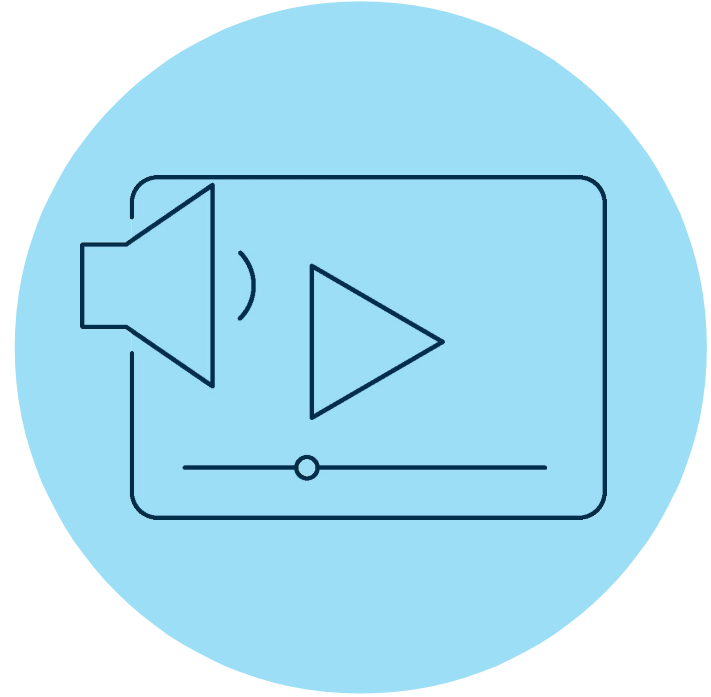


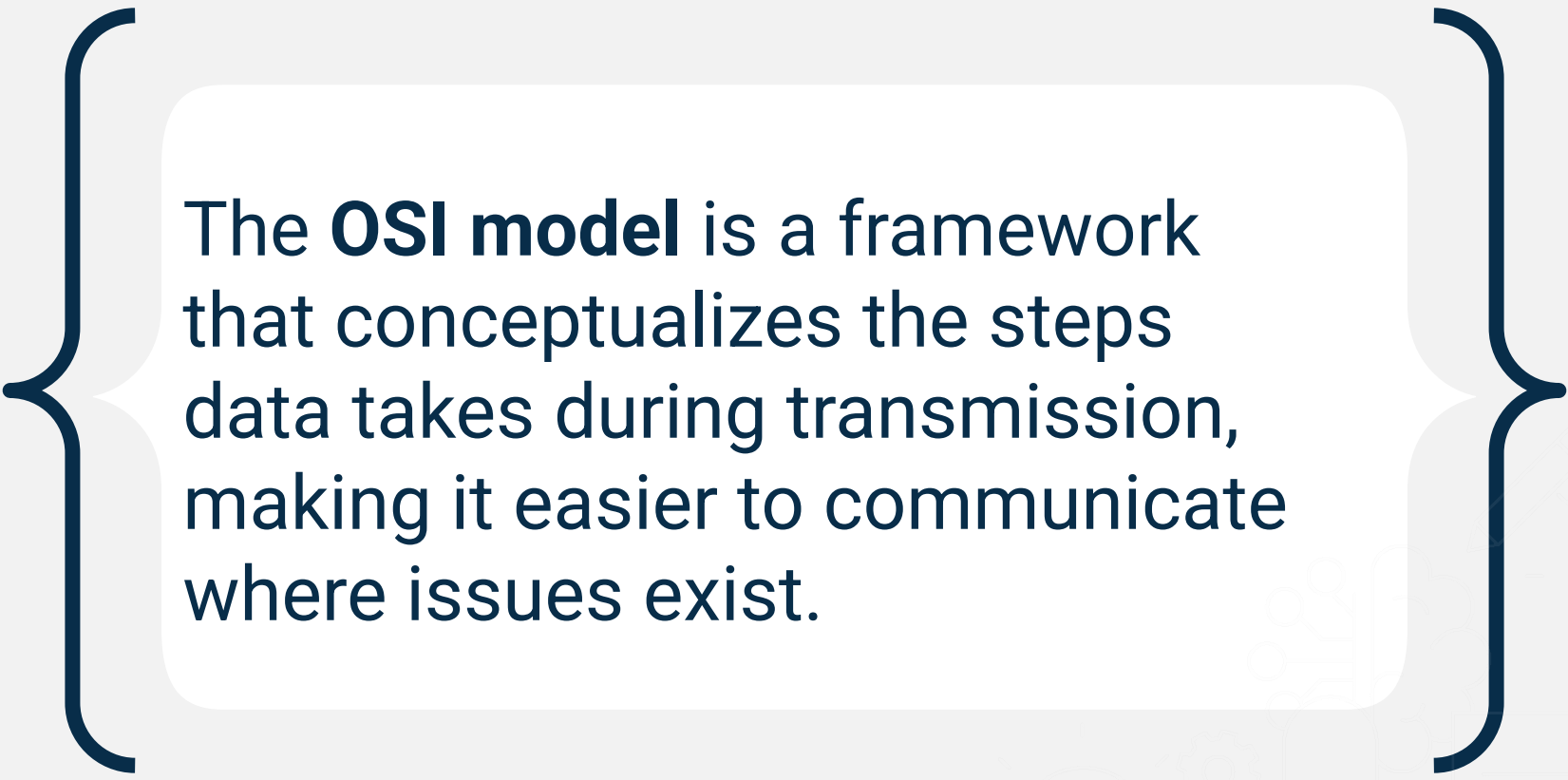


Since many of these steps are common across different scenarios, many protocols often perform the same “**general step.**”

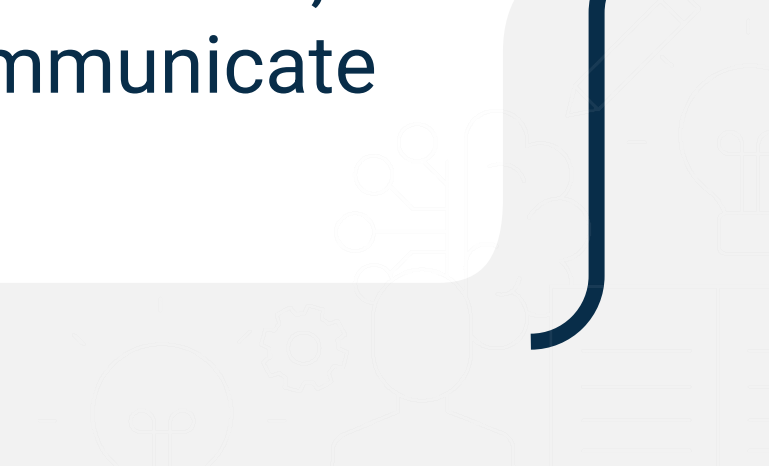
TCP and **UDP** are both used to transport data.

TCP is used to transport data from an email application, while UDP is used to transport data from a video streaming application.





The **OSI model** is a framework that conceptualizes the steps data takes during transmission, making it easier to communicate where issues exist.

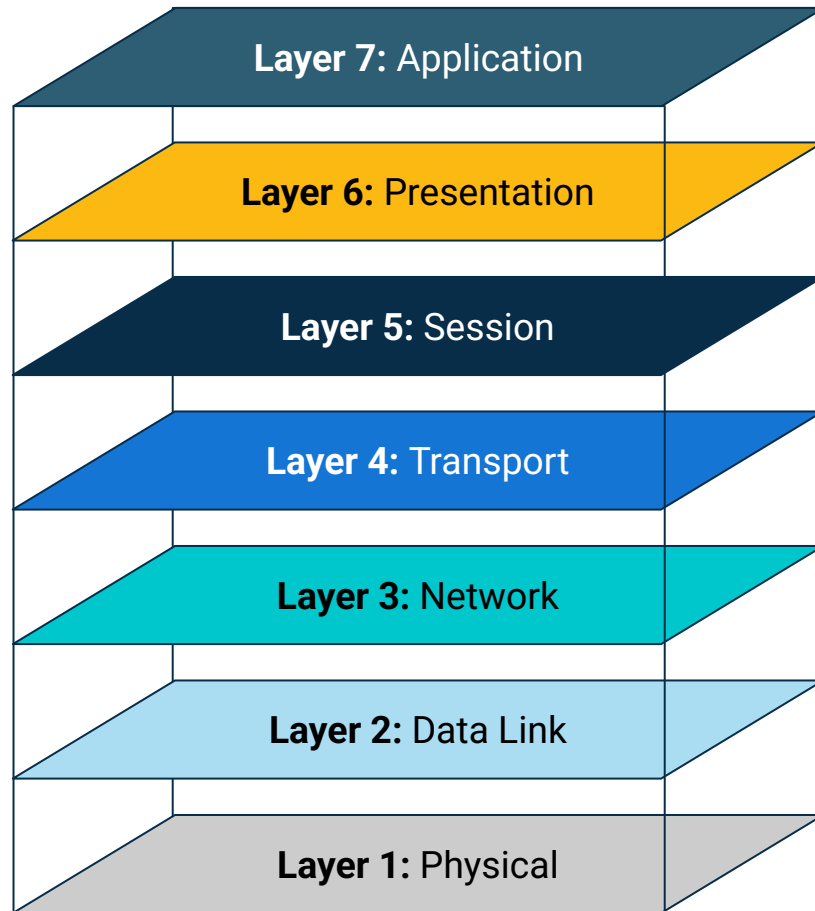


OSI Model Framework

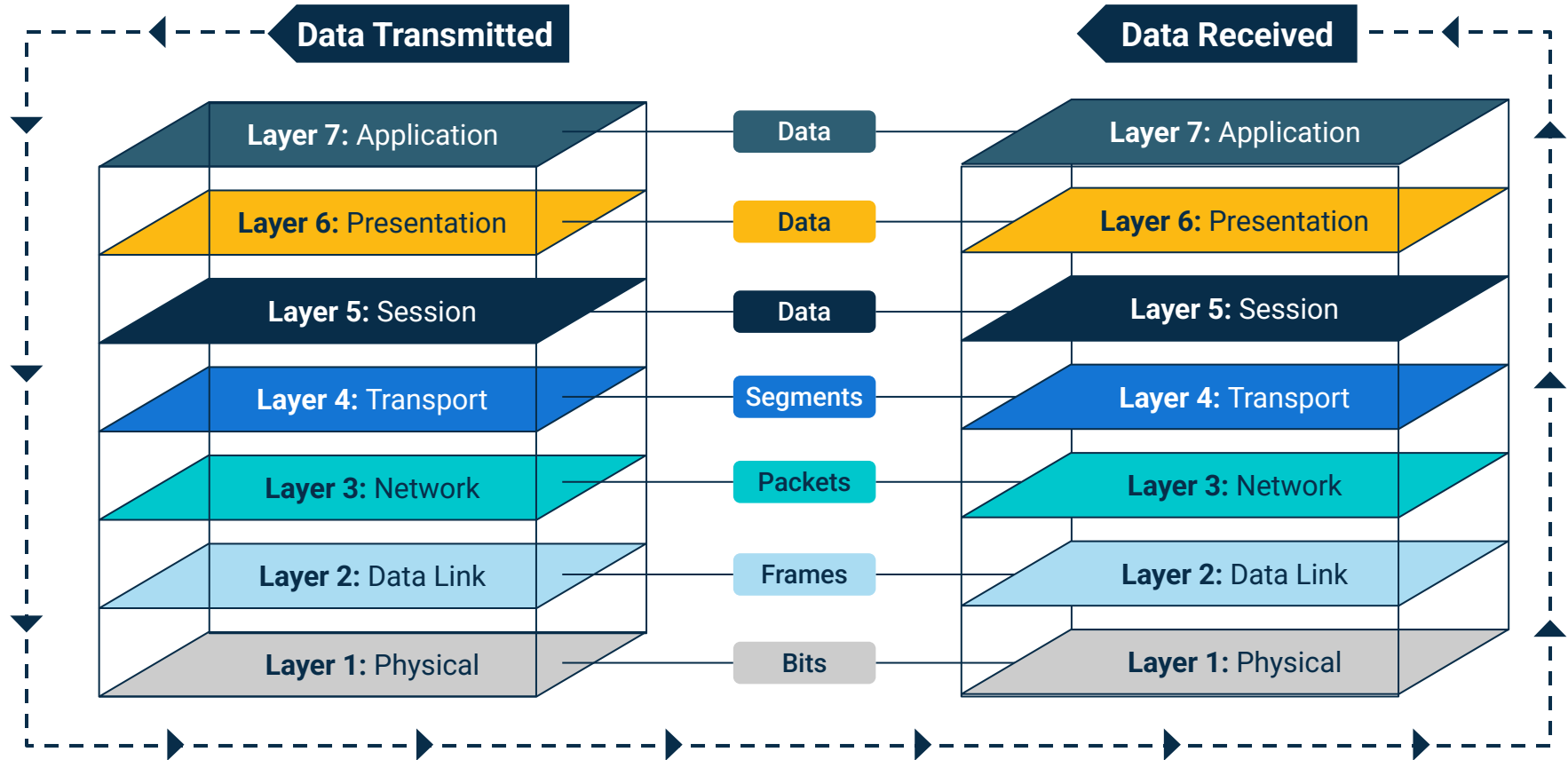
7 Layer conceptual framework

Allows security analysts to better understand how communication works on a network by detailing:

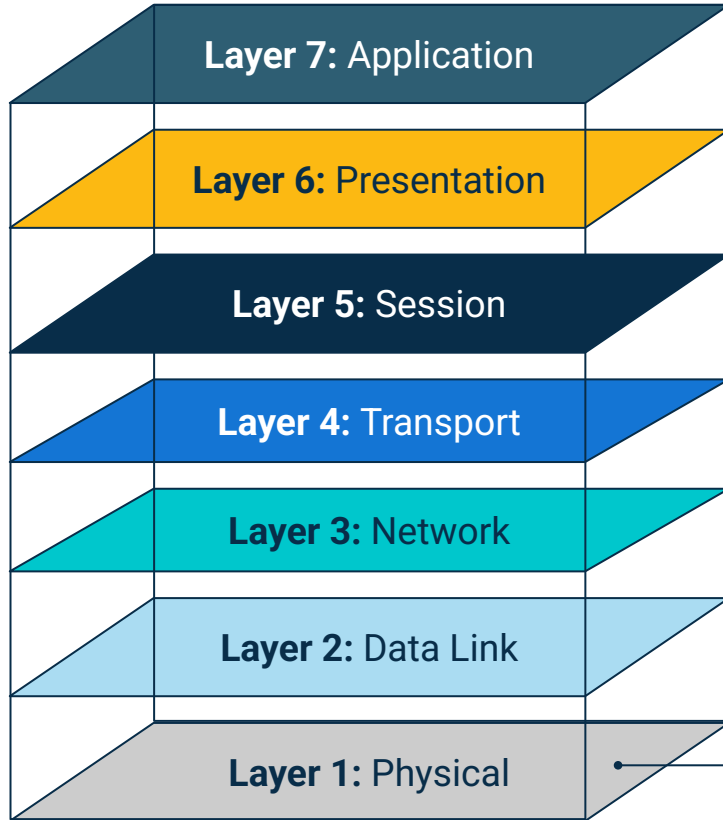
- Processes
- Devices
- Protocols



OSI Model



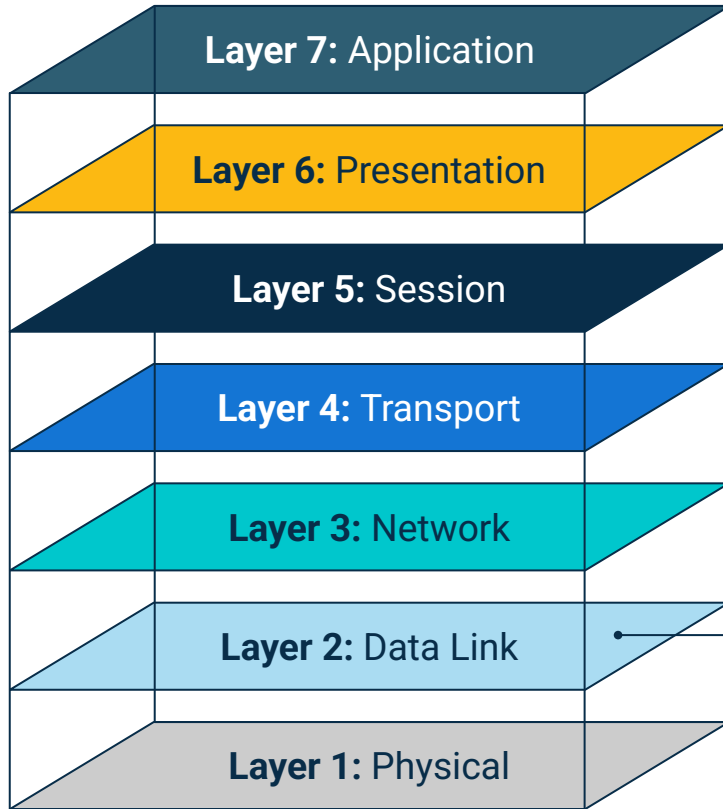
Layer 1: Physical



The **Physical layer** is responsible for transmitting binary data through a physical medium.

It handles how data is physically encoded and decoded.

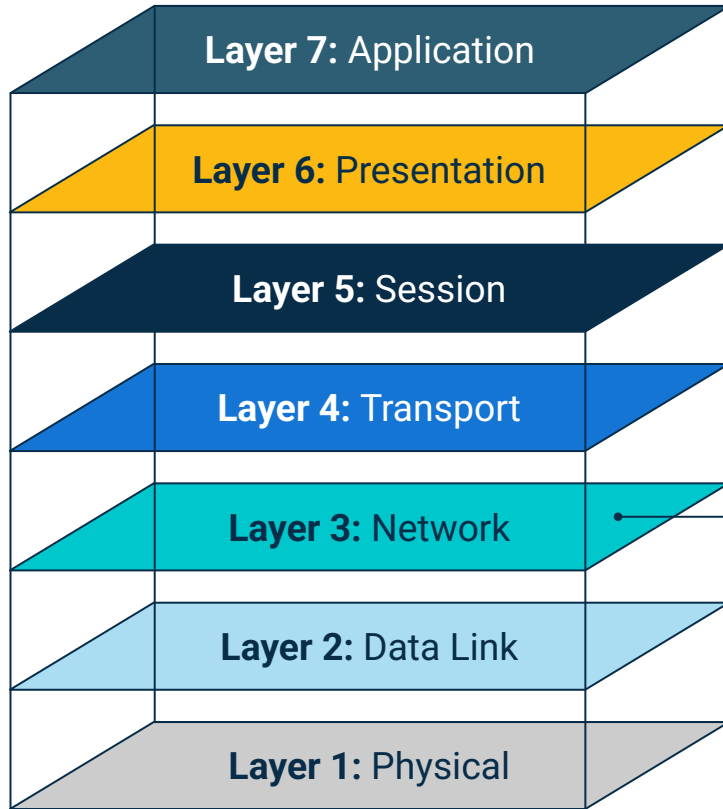
Layer 2: Data Link



The **Data Link** layer establishes links between nodes.

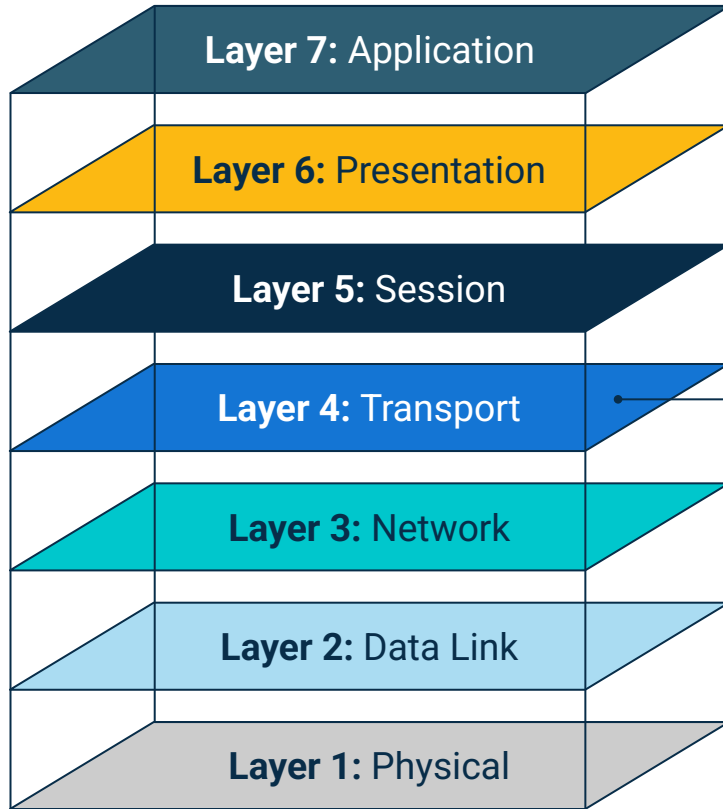
It also ensures data gets to its final destination without corruption, thus protecting data integrity.

Layer 3: Network



The **Network layer** routes data through physical networks using an IP address, deciding which physical path the data will take and ensuring it gets to the correct destination.

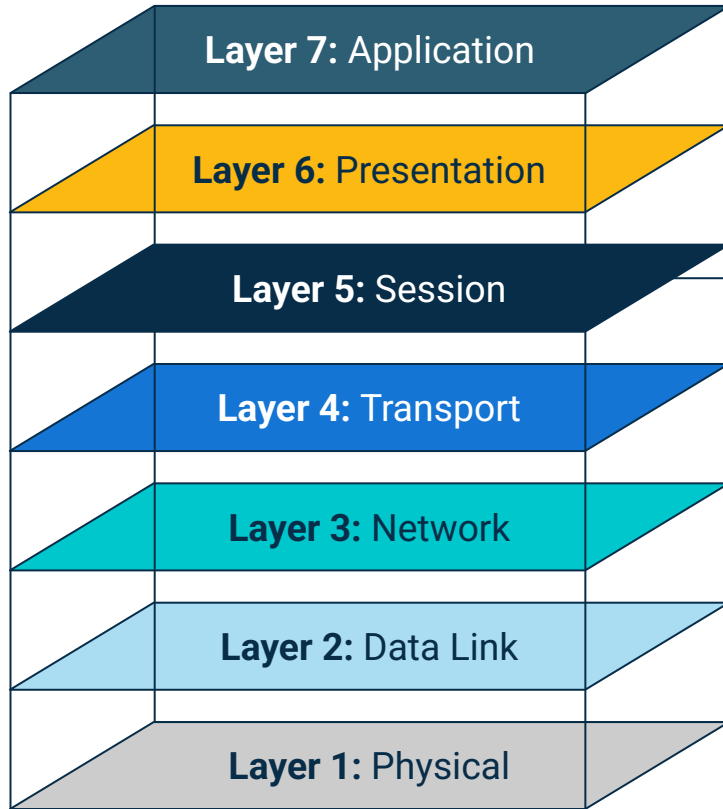
Layer 4: Transport



The **Transport layer** is responsible for actually transmitting data across the network.

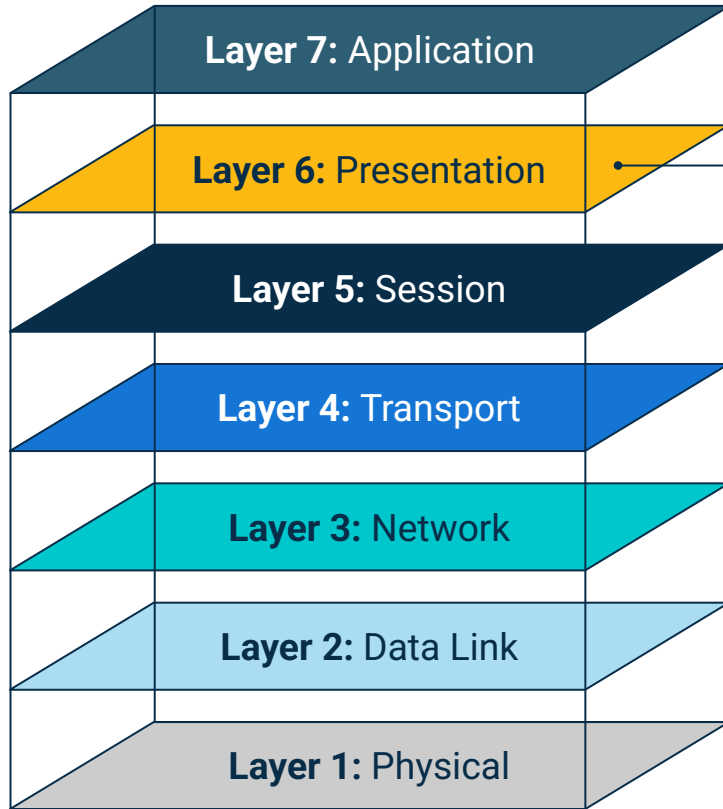
It puts data onto the network, and assigns source and destination ports.

Layer 5: Session



The **Session layer** manages connections between ports on computers and handles data flow.

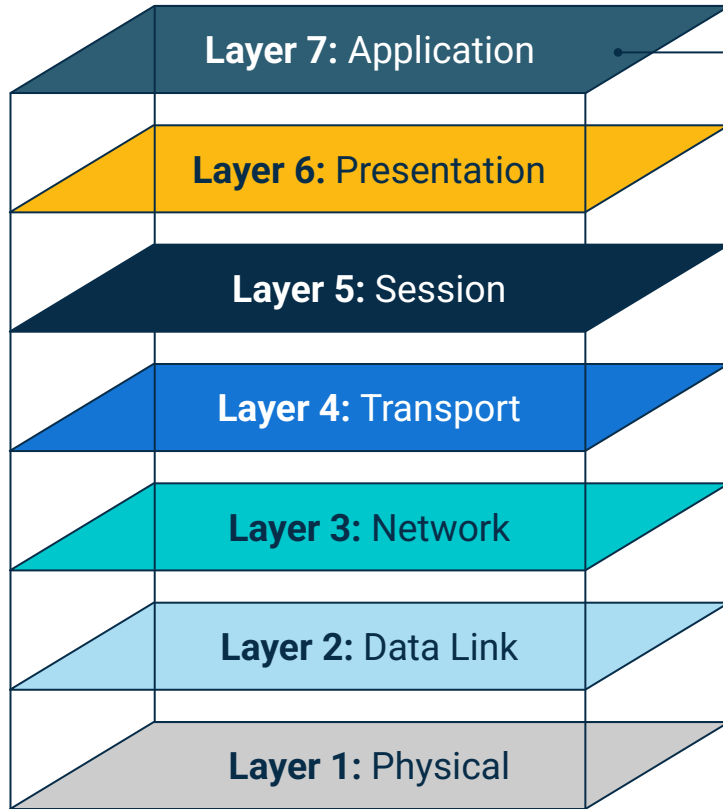
Layer 6: Presentation



The **Presentation layer** is the translator for the network.

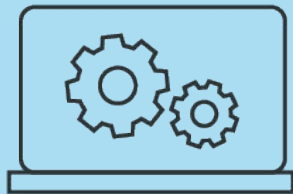
It formats data to be presented to the Application layer. It handles data representation, decryption, encryption, character-set translation, and conversion.

Layer 7: Application

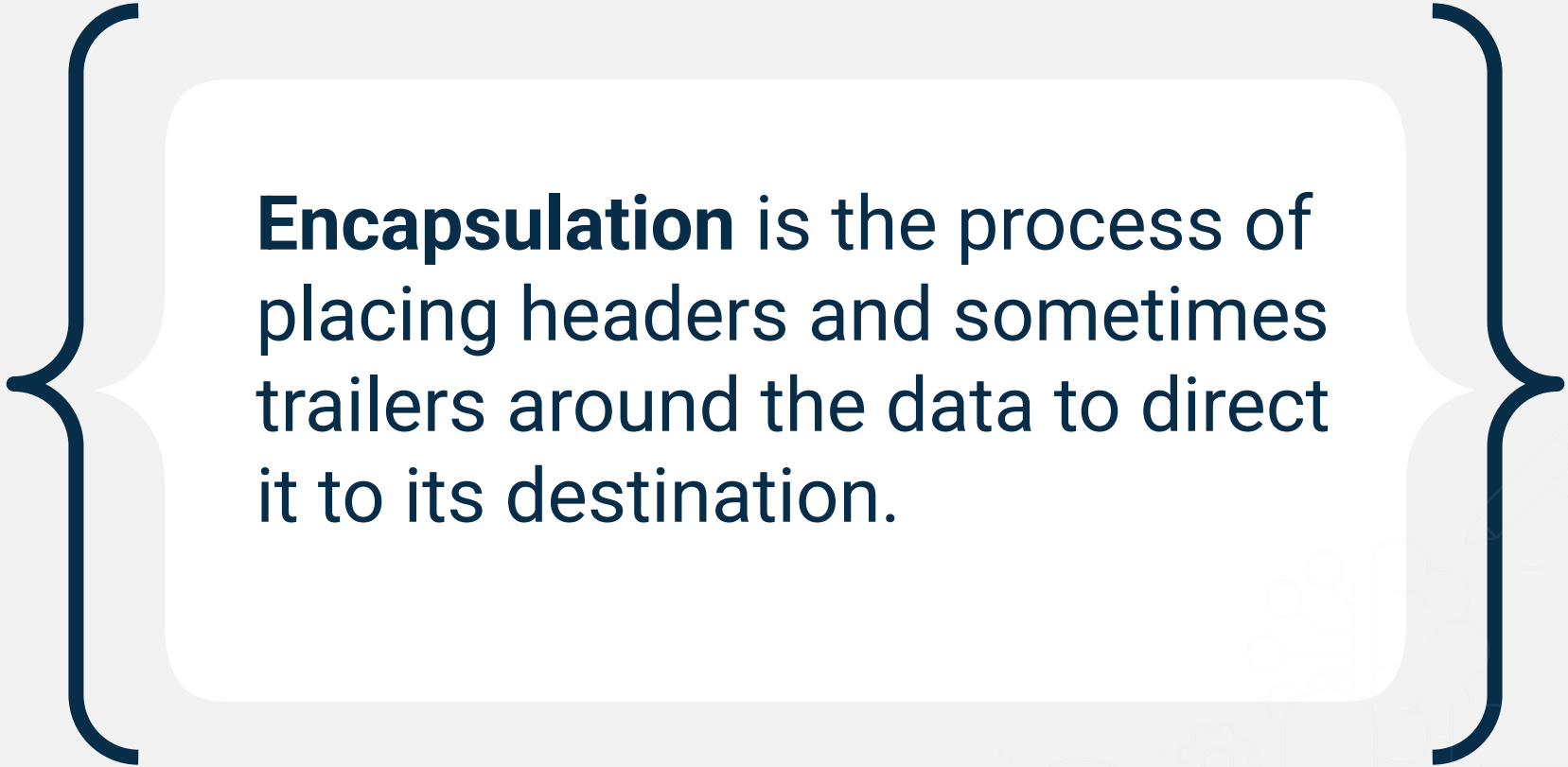


The **Application layer** is responsible for representing data in a way the consuming application can understand.

This is the layer a user interacts with, such as a web or email application.



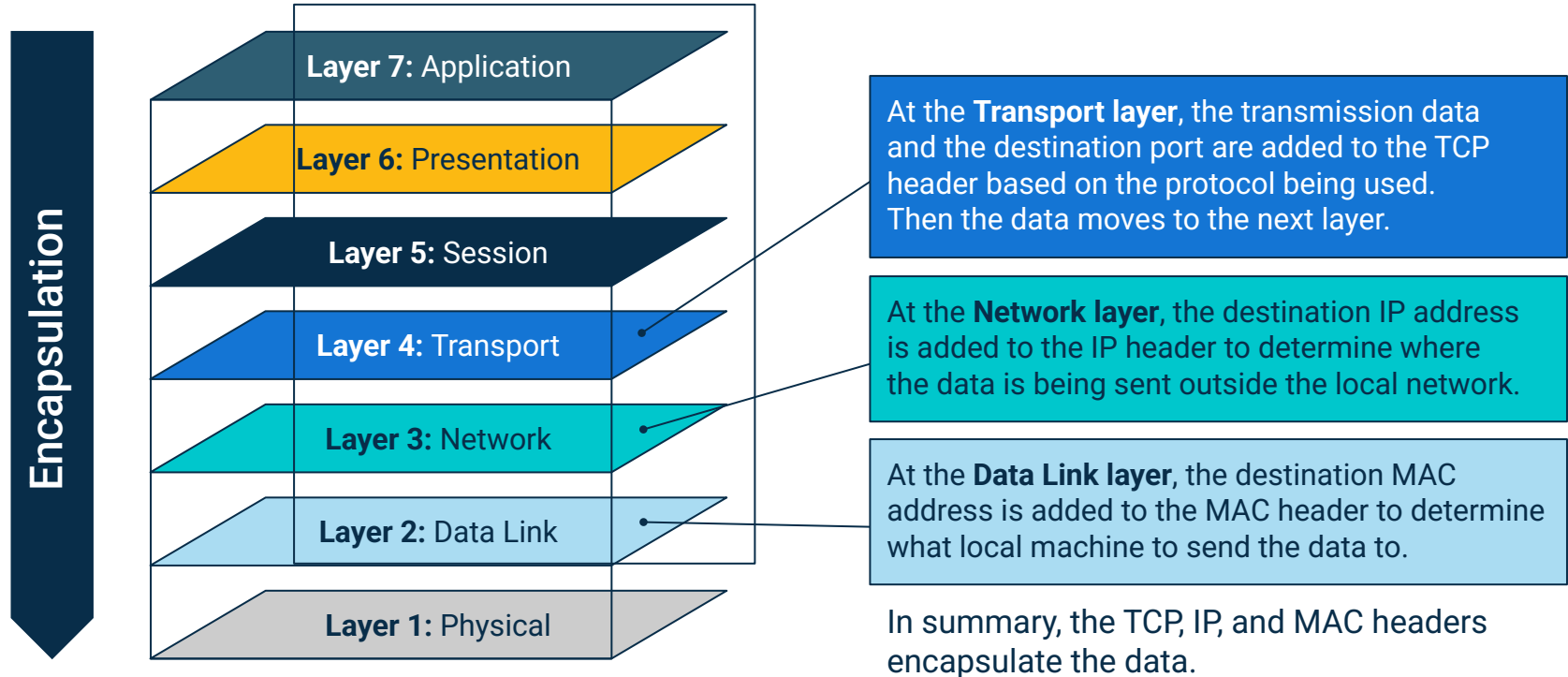
Encapsulation and Decapsulation

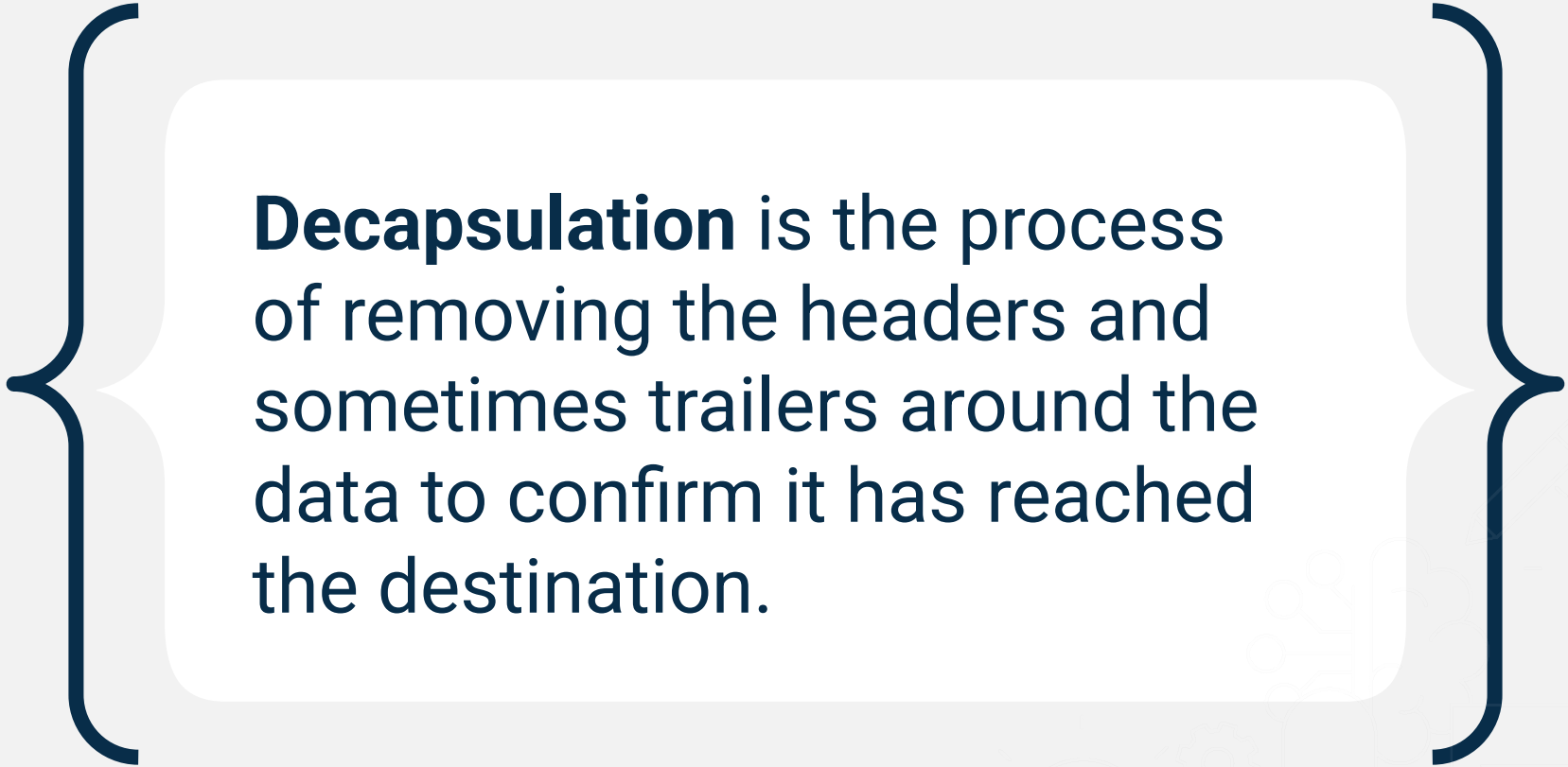


Encapsulation is the process of placing headers and sometimes trailers around the data to direct it to its destination.

Encapsulation

As data moves through the layers, starting from Layer 7 and ending at Layer 1, the data is **encapsulated**.

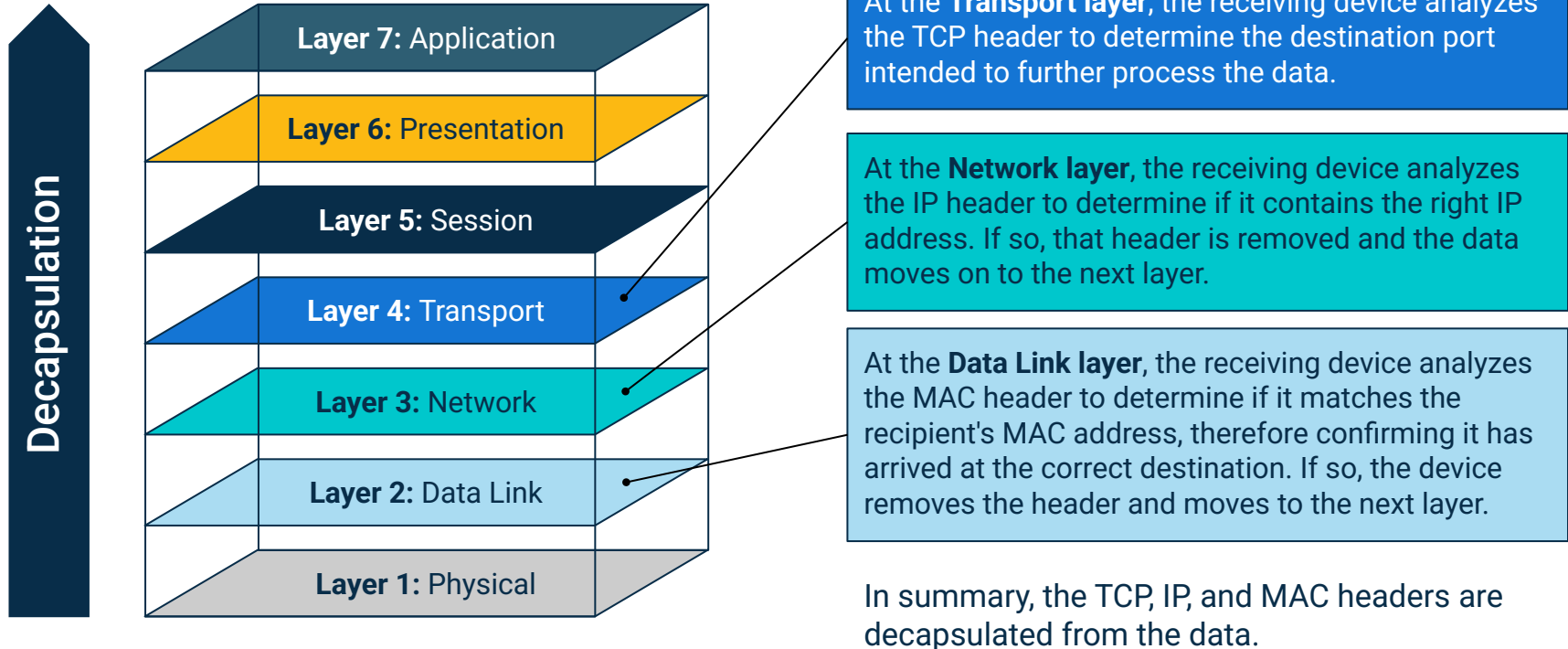




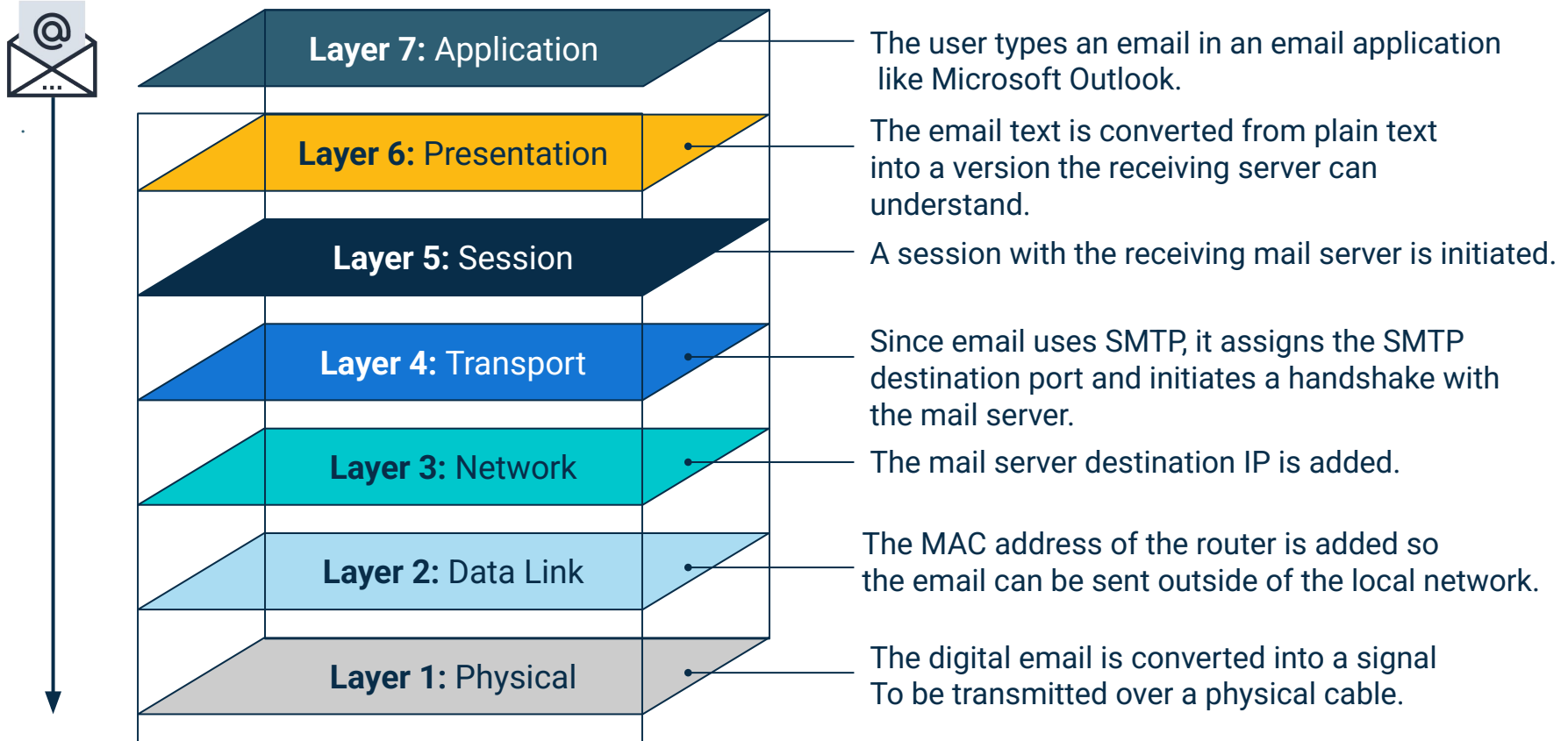
Decapsulation is the process of removing the headers and sometimes trailers around the data to confirm it has reached the destination.

Decapsulation

The following examples of **decapsulation** occur as data is moving across layers in reverse order:



An Email Moving Through the OSI Layers



OSI in a Security Context

The OSI model:

Example

Helps us easily understand new protocols.

If you find out that NetBIOS is a Layer 5 protocol, you immediately know that it's involved in managing user sessions, even if you've never heard of NetBIOS before.

Helps determine where problems in the network are occurring, even if we don't have full knowledge of the issue.

If you realize you're having a Layer 3 issue, you know you should start investigating your routers, even if you don't know exactly what the problem is.

Makes it easier to communicate where a security attack has occurred and what should be done.

If you know an SQL injection attack is occurring, you can explain to your management that you need a Layer 7 web application firewall to identify and mitigate the attack.



Activity

OSI layers activity

In this activity, you will continue to play the role of a security analyst at Acme Corp.

- Your manager provides you with a list of several suspicious network-related activities that have recently occurred at Acme Corp. Your manager believes Sally Stealer may be connected.
- Your task is to document at which OSI layer each of these situations occurred to help the networking team pull the necessary logs for investigation.

Suggested Time
15 minutes



Time's Up!
Let's review



Questions?





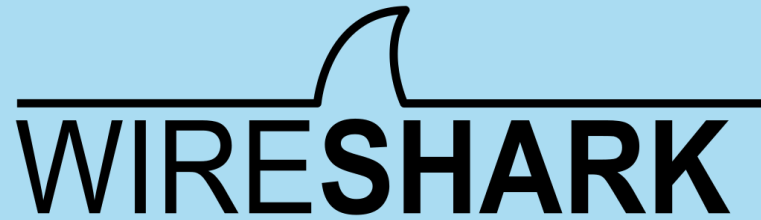
Let's Recap



Recap

In the last section, we discussed:

- 1 Networks communicate with sequences of binary data called **packets**.
- 2 Each packet contains fields such as the address of its origin, the address of its destination, and the information-related packets being sent.
- 3 These work a lot like the post office, except billions of packets are transferred every day and most packets take less than a few seconds to reach their destination.
- 4 Communicating over a network is not entirely safe as these packets can be intercepted and analyzed by other users on the network.
- 5 Cybersecurity professionals need to be able to see who is on a network and what they are doing. In other words, we have to be spies—at least a little bit.



Introduction to **Wireshark**

Introduction to Wireshark

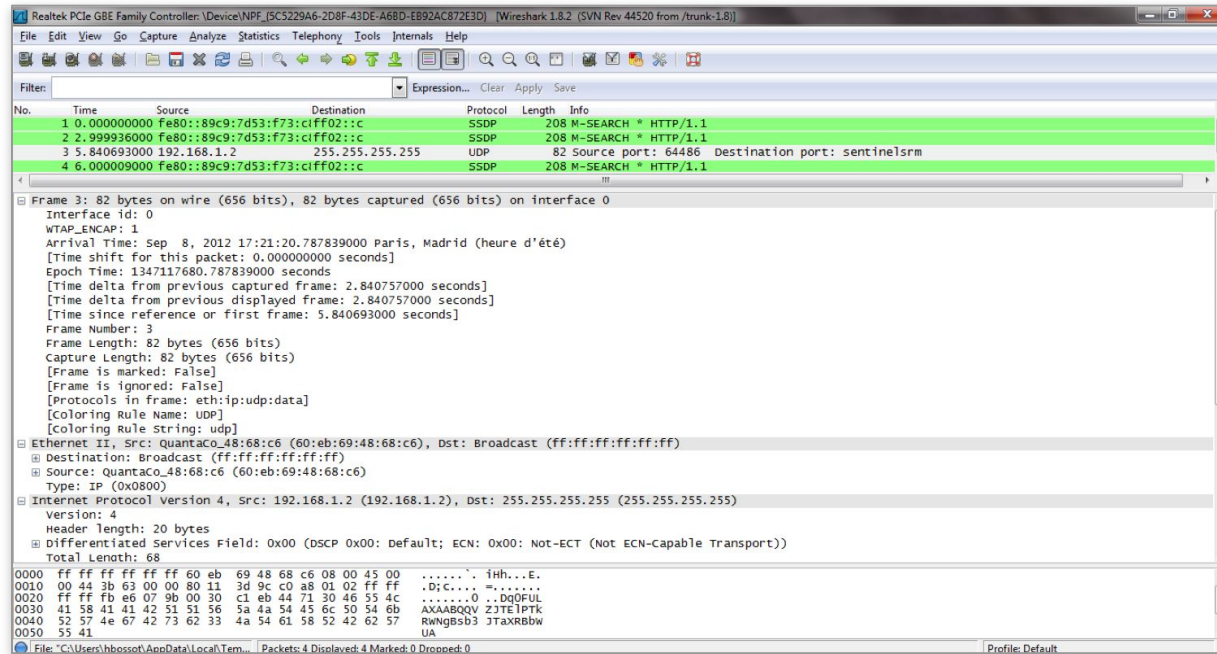
Real-time network monitoring tool

- Wireshark does this analysis by inspecting individual packets.
- Multiple packets collected into a file by Wireshark is called a **packet capture**.
- These have file extensions such as **.cap**, **.pcap**, and **.pcapng**.



Capturing Packets

In these packet captures, Wireshark collects and analyzes the kinds of websites and webpages individuals on the network are viewing, as well as the type of communication occurring.





Demonstration Scenario

tv-netflix-problems-2011-07-06.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/> Expression ... +

No.	Time	Source	Destination	Protocol	Length	Info
343	65.142415	192.168.0.21	174.129.249.228	TCP	66	40555 → 80 [ACK] Seq=1 Ack=1 Win=5888 Len=0 TSval=491519346 TSecr=551811827
344	65.142715	192.168.0.21	174.129.249.228	HTTP	253	GET /clients/netflix/flash/application.swf?flash_version=flash_lite_2.1&v=1.5&n=...
345	65.230738	174.129.249.228	192.168.0.21	TCP	66	80 → 40555 [ACK] Seq=1 Ack=188 Win=6864 Len=0 TSval=551811850 TSecr=491519347
346	65.240742	174.129.249.228	192.168.0.21	HTTP	828	HTTP/1.1 302 Moved Temporarily
347	65.241592	192.168.0.21	174.129.249.228	TCP	66	40555 → 80 [ACK] Seq=188 Ack=763 Win=7424 Len=0 TSval=491519446 TSecr=551811852
348	65.242532	192.168.0.21	192.168.0.1	DNS	77	Standard query 0x2188 A cdn-0.nflximg.com
349	65.276870	192.168.0.1	192.168.0.21	DNS	489	Standard query response 0x2188 A cdn-0.nflximg.com CNAME images.netflix.com.edge...
350	65.277992	192.168.0.21	63.80.242.48	TCP	74	37063 → 80 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM=1 TSval=491519482 TSecr=...
351	65.297757	63.80.242.48	192.168.0.21	TCP	74	80 → 37063 [SYN, ACK] Seq=0 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM=1 TSval=3295...
352	65.298396	192.168.0.21	63.80.242.48	TCP	66	37063 → 80 [ACK] Seq=1 Ack=1 Win=5888 Len=0 TSval=491519502 TSecr=3295534130
353	65.298687	192.168.0.21	63.80.242.48	HTTP	153	GET /us/nrd/clients/flash/814540.bun HTTP/1.1
354	65.318730	63.80.242.48	192.168.0.21	TCP	66	80 → 37063 [ACK] Seq=1 Ack=88 Win=5792 Len=0 TSval=3295534151 TSecr=491519503
355	65.321733	63.80.242.48	192.168.0.21	TCP	1514	[TCP segment of a reassembled PDU]

> Frame 349: 489 bytes on wire (3912 bits), 489 bytes captured (3912 bits)

> Ethernet II, Src: Globalelec_00:3b:0a (f0:ad:4e:00:3b:0a), Dst: Vizio_14:8a:e1 (00:19:9d:14:8a:e1)

> Internet Protocol Version 4, Src: 192.168.0.1, Dst: 192.168.0.21

> User Datagram Protocol, Src Port: 53 (53), Dst Port: 34036 (34036)

> Domain Name System (response)

Request In: 348

[Time: 0.03433000 seconds]

Transaction ID: 0x2188

Flags: 0x8180 Standard query response, No error

Questions: 1

Answer RRs: 4

Authority RRs: 9

Additional RRs: 9

Queries

> cdn-0.nflximg.com: type A, class IN

Answers

> Authoritative nameservers

0020 00 15 00 35 84 f4 01 c7 83 3f 21 88 81 00 00 01 ...S....?.....

0030 00 04 00 09 00 09 05 63 64 6e 2d 30 07 6e 66 6cc dn-0.nfl

0040 78 69 6d 67 83 63 6f 6d 00 00 01 00 01 c0 0c 00 ximg.com.....

0050 05 00 01 00 00 05 29 00 22 06 69 6d 61 67 65 73 ".images

0060 07 6e 65 74 66 6c 69 78 83 6f 6d 69 65 64 67 .netflix.com.edg

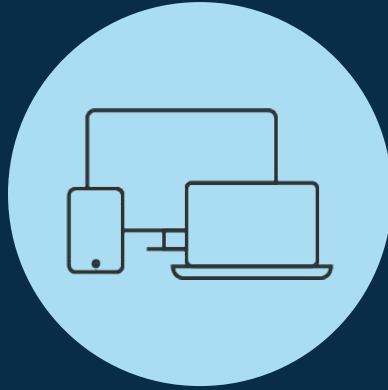
0070 65 73 75 69 74 65 03 6e 65 74 00 c0 2f 00 05 00 esuite.n et./...

Identification of transaction (dns.id), 2 bytes

Packets: 10299 - Displayed: 10299 (100.0%) - Load time: 0:0.182 | Profile: Default

Your manager tasks you with analyzing web traffic to determine which source ports your system is using for HTTP requests.

- They want to make sure these aren't being blocked by your firewall.
- Your manager provides you with a capture of the logs you need to analyze.



Instructor Demonstration

Wireshark



Activity

Capture packets

In this activity, you will continue to play the role of a security analyst at Acme Corp.

- Your task is to configure your Wireshark application with the five requested configuration settings provided by your manager. These settings will help you analyze Sally's network traffic, which we will capture in the final activity.

Suggested Time
15 minutes



Time's Up!
Let's review



Questions?





Analyze **HTTP** Data

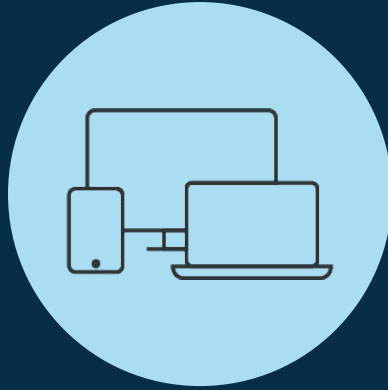


Demonstration Scenario

Your manager asks you to see if new employee Michael is, in fact, working hard on their first day of work.

- You can ask Michael if you could view their browser history, but this might not provide sufficient or accurate data—they could have cleared their browser history or used more than one browser.
- Fortunately, the networking team has a packet capture of Michael's web traffic, and you can use Wireshark to easily analyze the following:
 - **Which websites Michael visited**
 - **If any communications were sent from these websites**





Instructor Demonstration

Analyze HTTP data



Questions?





Activity

Analyze HTTP data activity

In this activity, you will continue to play the role of a security analyst at Acme Corp.

- Your task is to analyze the web traffic to determine what websites and webpages Sally accessed. You will also interrogate the logs to see if Sally sent any communications to WidgetCorp.

Suggested Time
15 minutes



Time's Up!
Let's review



Questions?





Next Class

Follow data through layers 2, 3, and 4



Thank you